

Postdoc - Robust optimisation of bifurcation diagrams for flutter design

General informations

Supervised by

- E. Denimal Goy and P.M. Congedo (Inria); Platon Inria project-team; Center for Applied Mathematics (Ecole Polytechnique)
- B. Chouvion (Ecole de l'Air et de l'Espace); CREA

Duration: 18 months

Starting date: no later than January 2027

Location: Inria Saclay, 1 rue Honoré d'Estienne d'Orves, 91120 Palaiseau, FRANCE

Salary: gross monthly salary of about 2700€

Funding: ANR FlexHALE, consortium of several French labs

Project description and objectives

The optimisation and the design of the dynamic behaviour of mechanical structures play a key role in many industries to meet stringent environmental and performance requirements. The consideration of the non-linearities in such structures is essential. These non-linearities are at the origin of numerous and complex behaviours such as the softening or stiffening of the resonance peak, the existence of multiple dynamic solutions and the appearance of bifurcations in the dynamic behaviour. Bifurcations represent a stability limit in the parameter space characterised by a qualitative and quantitative change in the dynamics of the system (e.g. number and type of responses).

For example, in the development of HALE (High Altitude Long Endurance) or HAPS (High Altitude Pseudo-Satellite) drones, such as the HELIOS drone developed by NASA, the control of aeroelastic instability phenomena is a major challenge. Among these, aeroelastic flutter—resulting from the coupling between structural dynamics and aerodynamic forces—can lead to severe structural failures if not predicted with sufficient accuracy. Classic methods for predicting the critical flutter speed typically rely on deterministic models [5], whereas in practice, numerous sources of uncertainty exist, particularly related to aerodynamic properties, structural mechanical characteristics, or operational conditions. Explicitly accounting for these uncertainties is therefore crucial to identify reliable and robust designs [6].

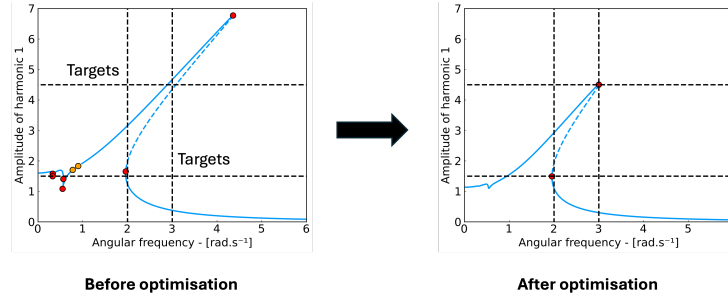


Figure 1: Illustration of the optimisation of bifurcation diagrams - Frequency Response Curves before (left) and after (right) optimisation in blue, bifurcation points in red, and frequency and amplitude targets with black lines

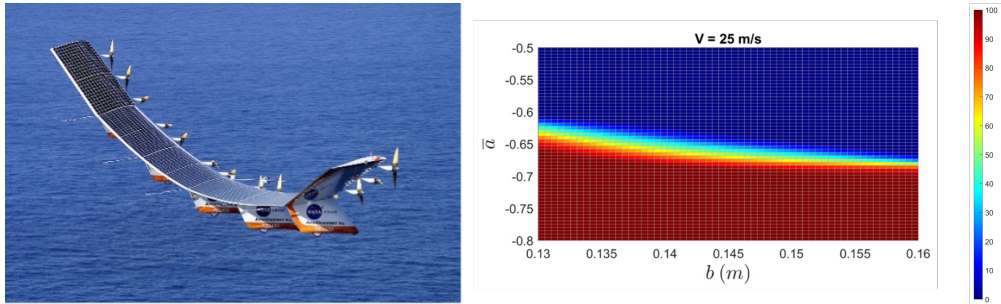


Figure 2: HELIOS Drone (left) and % of occurrence of flutter for an airfoil under uncertain conditions

Recent works from the team have focused on the deterministic optimisation of mechanical structures to reach desired bifurcation behaviours [2, 1], see Figure . However, numerous uncertainties are present, either from the aerodynamic properties or from mechanical properties. The impact of those uncertainties is critical as the system stability can be impacted [4, 3]. Their consideration from the structural optimisation is crucial to ensure the robustness and reliability of the mechanical design.

The objective of the postdoc is to develop robust optimisation methods for bifurcation diagrams. The aim is to combine technics for the analysis of bifurcation of optimization and of uncertainty quantification. Large parametric variations will be considered in the optimisation, leading to large structural variations and so a large range of dynamic behaviours. The bifurcation analysis as well as uncertainty propagation steps are numerically expensive and surrogate-based strategies will be investigated in order to reduce the numerical cost. Three main objectives have been identified for the postdoc:

- The development of the uncertainty propagation methods for the characterisation of bifurcation behaviour of stochastic nonlinear dynamic systems.
- The development of robust optimisation methods for bifurcation diagrams
- The development of methods able to deal with real-world scenarios, and more particularly on the test case of a HALE drone for flutter mitigation.

The person recruited will have to numerically implement, test and compare the different identified approaches developed during the postdoc.

Supervision

The postdoc will be supervised by E. Denimal Goy and P.M. Congedo, experts in uncertainty quantification methods for engineering applications. He/She will be also supervised by B. Chouvion from CREA/Ecole de l'Air et de l'Espace, where he has developed a physical solver for flutter calculation and characterization for mechanical structures with geometric nonlinearities and aerodynamic coupling.

The work will be conducted in the Platon team, a joint research group between Ecole Polytechnique and CNRS, hosted by the Center for Applied Mathematics (CMAP) of École Polytechnique. The Platon project-team focuses on developing innovative methods and algorithms for uncertainty management in numerical models, including advanced calibration strategies from data (observations, measurements, other model predictions) and uncertainty reduction.

Required skills, duration and application procedure

Candidates must hold a PhD in mechanical engineering, applied mathematics or a related discipline with background in at least one of these fields: non-linear dynamics, uncertainty quantification, robust optimisation or related fields. In particular, candidates must be proficient scientific computing

Applicants should submit a detailed academic CV with history of scientific production, evaluation documents of their PhD if available and a cover letter detailing the knowledge, skills and experience you think make you the right candidate for the job. For further details, please contact E. Denimal Goy (enora.denimal-goy [at] inria.fr) and B. Chouvion (benjamin.chouvion [at] ecole-air.fr).

Applications must be submitted at this link.

References

- [1] A. Mélot, E. Denimal Goy, L. Renson, *Control of isolated response curves through optimization of codimension-1 singularities*, Computers & Structures, 2024, 299: 107394.
- [2] A. Mélot, E. Denimal, L. Renson, *Multi-parametric optimization of bifurcation structures*, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 2024, 480:2023050520230505
- [3] E. Denimal, J-J. Sinou, *Efficient parametric study of a stochastic airfoil system based on hybrid surrogate modelling with advanced automatic kriging construction*, European Journal of Mechanics-A/Solids, 2023, 99: 104926
- [4] E. Denimal, J-J. Sinou, S. Nacivet, *Influence of structural modifications of automotive brake systems for squeal events with kriging meta-modelling method*, Journal of Sound and Vibration, 2019, 463: 114938
- [5] R. Alcorta, B. Chouvion, G. Michon, O. Montagnier, *On the use of frictional dampers for flutter mitigation of a highly flexible wing*, International Journal of Non-Linear Mechanics, 2023, 156 :104515.

- [6] Razaaly, N. et Chouvion, B. *Quantile-Based Reliability-Constrained Optimization of a Nonlinear Absorber for Passive Aeroelastic Control under Aleatory Uncertainty*, Structural and Multidisciplinary Optimization, 2026.